

Title

Evaluating Multi-Sector Integration of Ecological Principles in School-Based Weight Management: A Mixed-Methods Analysis Protocol

Abstract

Childhood obesity prevention requires simultaneous intervention across multiple levels of influence including individual, family, school, and policy sectors. This study will evaluate a comprehensive ecological initiative implemented across 15 elementary schools serving urban populations. The initiative encompasses environmental modifications, curricula adjustment, family engagement programming, and municipal policy coordination targeting food and activity environments. Analysis will examine the extent to which implemented strategies reflect core ecological principles such as social-ecological models and reciprocal determinism between behaviour and environmental context. Qualitative and quantitative coding frameworks will assess which theoretical constructs underpin specific intervention components. The research will map implementation fidelity and identify which strategy types (environmental vs. educational vs. policy) are prioritised in resource allocation. This evaluation aims to clarify which combinations of ecological and behavioural science principles appear most effective in real-world school settings, generating actionable insights for replication.